

GCP Database Migration Guide

Current Status

Completed Steps:

1.  **Schema Updated** - `prisma/schema.prisma` updated with 16 new models
2.  **Migration Files Created** - Located at `prisma/migrations/20251106012548_complete_saas_implementation/`
3.  **DATABASE_URL Configured** - `.env` file contains properly encoded connection string

Migration File Details:

- **Location:** `prisma/migrations/20251106012548_complete_saas_implementation/migration.sql`
 - **Size:** 823 lines of SQL
 - **Content:** All tables, enums, indexes, and foreign key constraints
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Network Access Issue

Problem:

The GCP Cloud SQL instance at `136.116.45.158:5432` is **not accessible** from external networks. This is expected behavior for Cloud SQL security.

Error Encountered:

```
Can't reach database server at `136.116.45.158:5432`
```

Solution: Configure GCP Cloud SQL Network Access

Option 1: Add Authorized Networks (Recommended for Development)

1. **Open Google Cloud Console**
 - Navigate to: `https://console.cloud.google.com/sql/instances`
 - Select your project: `crtlpyme-chile-2025`
 - Click on your instance: `crtlpyme-db`
2. **Configure Connections**
 - Go to "Connections" tab
 - Click "Networking"
 - Under "Authorized networks", click "Add network"
 - Add your IP address or `0.0.0.0/0` (for testing only - not recommended for production)
 - Click "Save"

3. Wait for Configuration

- Changes take 1-2 minutes to apply

Option 2: Use Cloud SQL Proxy (Recommended for Production)

1. Install Cloud SQL Proxy:

```
bash
curl -o cloud-sql-proxy https://storage.googleapis.com/cloud-sql-connectors/cloud-sql-proxy/v2.14.2/cloud-sql-proxy.linux.amd64
chmod +x cloud-sql-proxy
```

2. Start Proxy (requires authentication):

```
bash
./cloud-sql-proxy crtlyme-chile-2025:southamerica-east1:crtlyme-db
```

3. Update DATABASE_URL in .env :

```
DATABASE_URL="postgresql://postgres:%7B%60kkE%5B8SyJ%40G_IyF@localhost:5432/crtlyme-db?schema=public"
```

Option 3: Use Private IP with VPN

For production environments, use Private IP connectivity with Cloud VPN or Interconnect.

Step 4: Apply Migrations to GCP Database

Once network access is configured, run these commands:

Method 1: Prisma Migrate Deploy (Recommended)

```
cd /home/ubuntu/github_repos/CRTLPyme

# Verify connection
npx prisma db pull --force

# Apply migrations
npx prisma migrate deploy

# Verify migrations applied
npx prisma migrate status
```

Method 2: Manual SQL Execution (Alternative)

If Prisma migrate fails, you can apply the SQL directly:

```
# Using psql
PGPASSWORD='{`kkE[8SyJ@G_IyF' psql -h 136.116.45.158 -U postgres -d crtlyme-db -f prisma/migrations/20251106012548_complete_saas_implementation/migration.sql
```

Step 5: Generate Prisma Client

After migrations are applied:

```
cd /home/ubuntu/github_repos/CRTLPyme
npx prisma generate
```

This generates the TypeScript client with all 16 new models.

Database Connection Details

Connection Information:

- **Host:** 136.116.45.158
- **Port:** 5432
- **Database:** crtlpyme-db
- **User:** postgres
- **Password:** {`kkE[8Sy]@G_IyF (special characters URL-encoded in .env)

Encoded DATABASE_URL:

```
postgresql://postgres:%7B%60kkE%5B8SyJ%40G_IyF@136.116.45.158:5432/crtlpyme-db?
schema=public
```

Character Encoding Reference:

- { → %7B
- ` → %60
- [→ %5B
- @ → %40

New Database Models (16 Total)

SaaS Administration (5 models):

1. **PlatformAdmin** - Super admins managing the platform
2. **PlatformAdminSession** - Admin authentication sessions
3. **TenantManagement** - Enhanced tenant management
4. **TenantActionLog** - Audit log for tenant actions
5. **SubscriptionMetric** - Analytics for subscription plans

Subscription Management (2 models):

1. **SubscriptionPlan** - Available subscription plans
2. **Subscription** - Active tenant subscriptions

Email System (3 models):

1. **EmailTemplate** - Reusable email templates
2. **EmailQueue** - Outgoing email queue
3. **EmailLog** - Email delivery tracking

Notification System (2 models):

1. **NotificationPreference** - User notification settings
2. **NotificationHistory** - Sent notifications log

Payment Processing (4 models):

1. **SubscriptionPayment** - Payment records
2. **PaymentWebhook** - Webhook event logs
3. **Refund** - Refund processing
4. **TransbankTransaction** - Transbank integration data

Verification Commands

After applying migrations, verify the schema:

```
# Check all tables
npx prisma db pull

# List all tables in database
PGPASSWORD='{`kkE[8SyJ@G_IyF' psql -h 136.116.45.158 -U postgres -d crtlpyme-db -c "\dt"

# Count tables
PGPASSWORD='{`kkE[8SyJ@G_IyF' psql -h 136.116.45.158 -U postgres -d crtlpyme-db -c "SELECT COUNT(*) FROM information_schema.tables WHERE table_schema = 'public';"
```

Troubleshooting

Issue: “Can’t reach database server”

Solution: Configure authorized networks in Cloud SQL (see above)

Issue: “Password authentication failed”

Solution: Verify password encoding in DATABASE_URL

Issue: “Database does not exist”

Solution: Create database:

```
CREATE DATABASE "crtlpyme-db";
```

Issue: “Migration already applied”

Solution: Check migration status:

```
npx prisma migrate status
```

Next Steps After Migration

1.  Verify all tables created
 2.  Generate Prisma Client
 3.  Test basic CRUD operations
 4.  Seed initial data (subscription plans, email templates)
 5.  Configure Cloud Build CI/CD
 6.  Set up automatic database backups in GCP
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Support

- **GCP Project:** crtlpyme-chile-2025
- **Database Instance:** crtlpyme-db
- **Region:** southamerica-east1
- **GitHub Repo:** <https://github.com/kbzas090/CRTLPyme>

For questions, check:

- [Prisma Migration Docs](https://www.prisma.io/docs/concepts/components/prisma-migrate) (<https://www.prisma.io/docs/concepts/components/prisma-migrate>)
- [Cloud SQL Connectivity](https://cloud.google.com/sql/docs/postgres/connect-overview) (<https://cloud.google.com/sql/docs/postgres/connect-overview>)